WATER QUALITY ANALYSIS AND POTABILITY PREDICTION

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What is Machine Learning?

Machine learning (ML) is a subset of artificial intelligence (AI) that enables systems to learn from data, identify patterns, and make decisions or predictions without being explicitly programmed for every task. Instead of hard-coded instructions, ML models use algorithms to analyze, learn from data, and improve their performance through experience.

Importance in real-world applications?

- Enables quick detection of unsafe water.
- Reduces Time and cost for manual testing.
- Helps improve public health monitoring.
- Identifies key factors affecting water quality.
- Scalable for large datasets and real-time systems.
- Models like Random – Forest handle complex patterns effectively.

Problem Statement

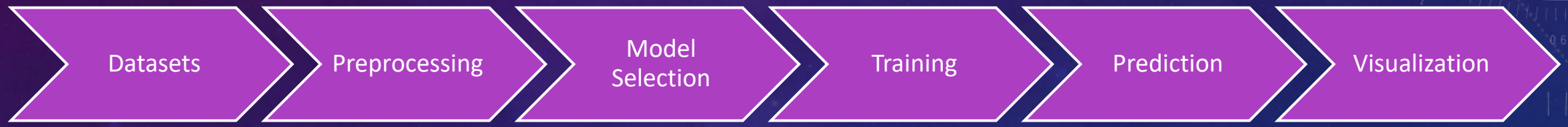
Ensuring safe drinking water is challenging due to pollution and limitations of traditional testing methods, which are time-consuming and costly. This project uses machine learning to analyze water quality parameters and predict potability, providing a fast and efficient solution using models like Random Forest.

- Water quality is critical for human health.
- Contaminated water can cause serious diseases.
- Manual testing is time-consuming and costly.
- Need for automated prediction system.

Objectives

- Analyze water quality dataset
- Perform Exploratory Data Analysis (EDA)
- Handle missing values
- Build a Machine Learning model
- Predict water potability
- Identify important features

System Overview



Dataset Description

- Dataset: Water potability dataset
- Contains chemical properties of water
- Target variable:
  - Potability(0=Not safe, 1=Safe)
- Key features:
  - pH
  - Hardness
  - Solids
  - Chloramines
  - Sulfate
  - Conductivity
  - Organic carbon
  - Trihalomethanes
  - Turbidity

EXPLORATORY DATA ANALYSIS (EDA)

- Univariate Analysis:
  - Distribution of pH, Hardness, Solids
- Bivariate Analysis:
  - Relationship with Potability
- Multivariate Analysis:
  - Correlation Heatmap

DATA PREPROCESSING

- Missing values found in dataset
- Handled using **Median Imputation**
- Feature Scaling using **Standard Scaler**
- Data split into:
    - Training set (80%)
    - Testing set (20%)

MACHINE LEARNING MODEL

- Model Used: **Random Forest Classifier**
- Ensemble learning technique
- Uses multiple decision trees
- Handles non-linear data effectively
- Robust and accurate

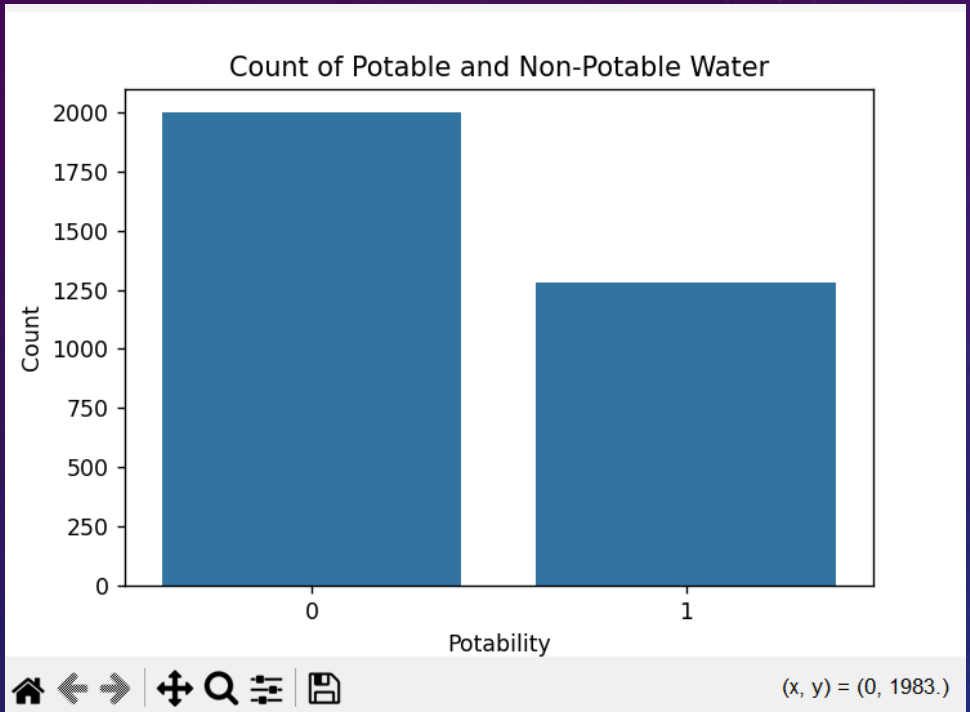
MODEL EVALUATION

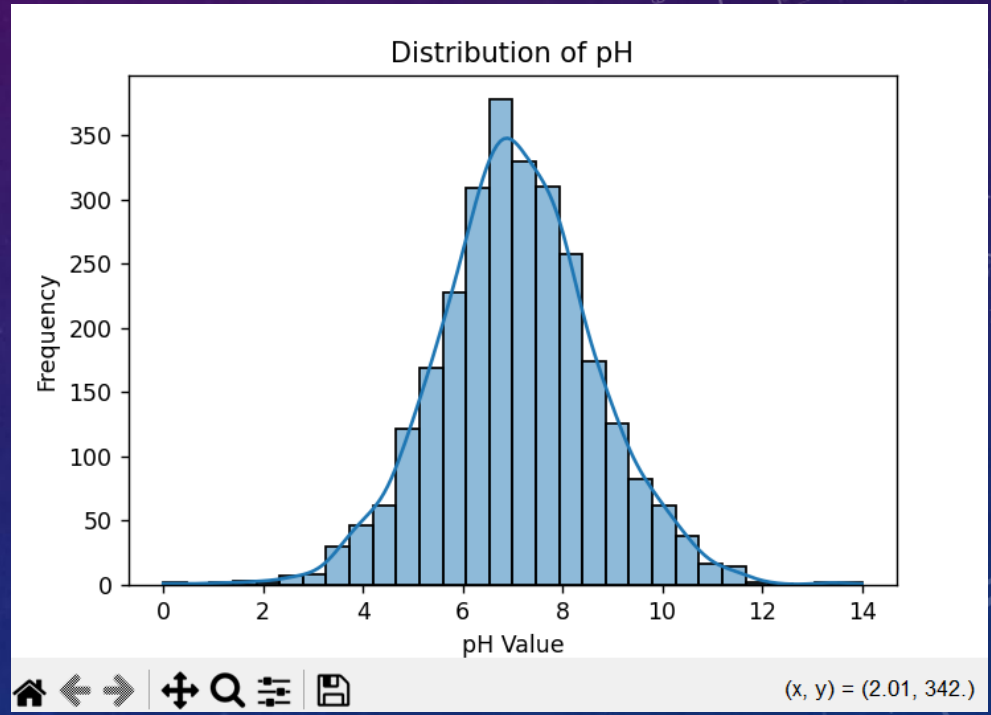
- Accuracy Score
- Confusion Matrix
- Classification Report:
    - Precision
    - Recall
    - F1-score

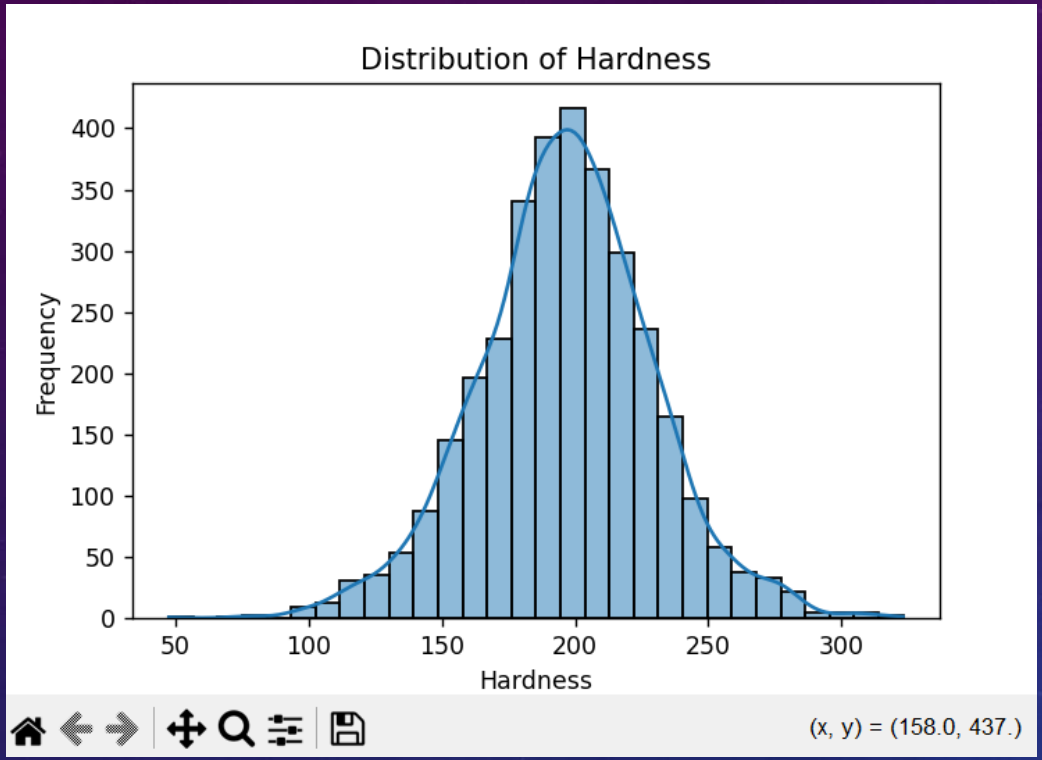
FEATURE IMPORTANCE & PCA

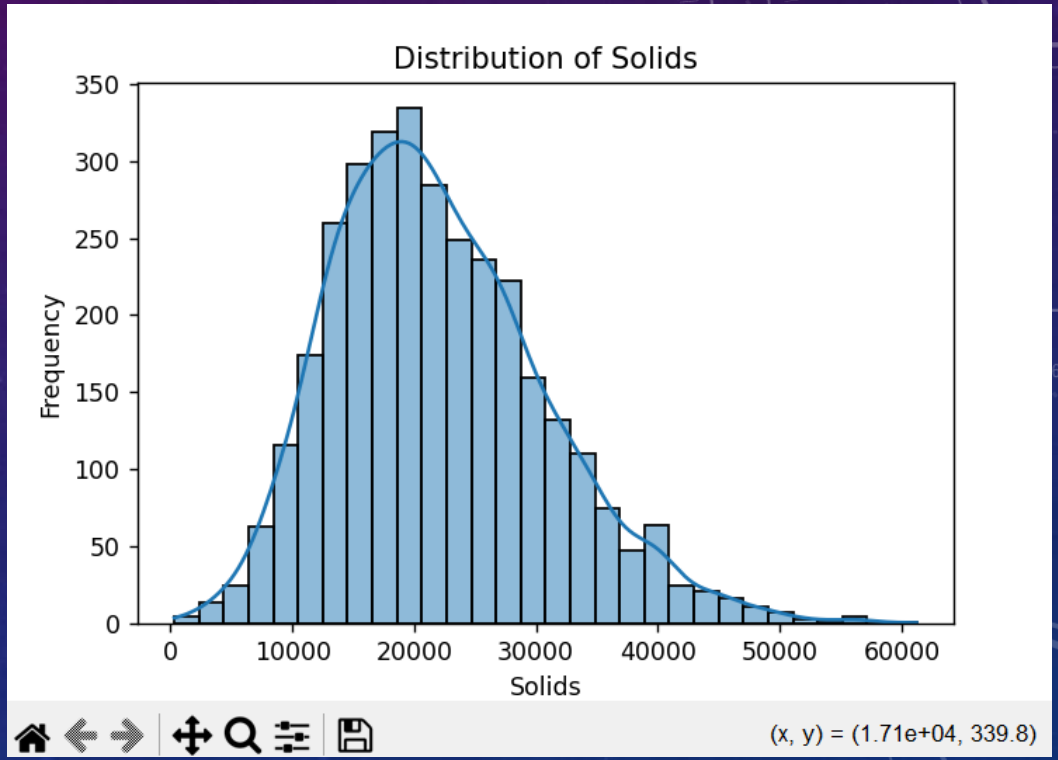
- Feature Importance:
  - Identifies most influential factors
- PCA (Principal Component Analysis):
  - Reduces dimensions to 2 components
  - Helps in visualization

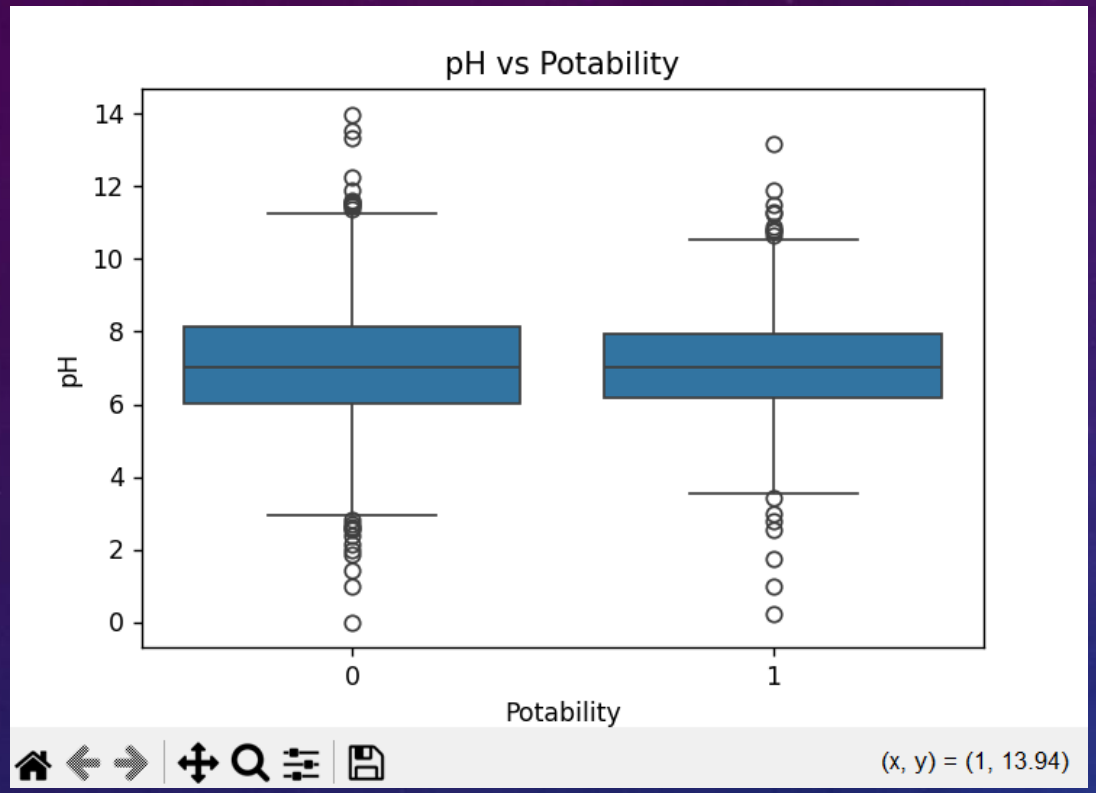
Outputs of our dataset

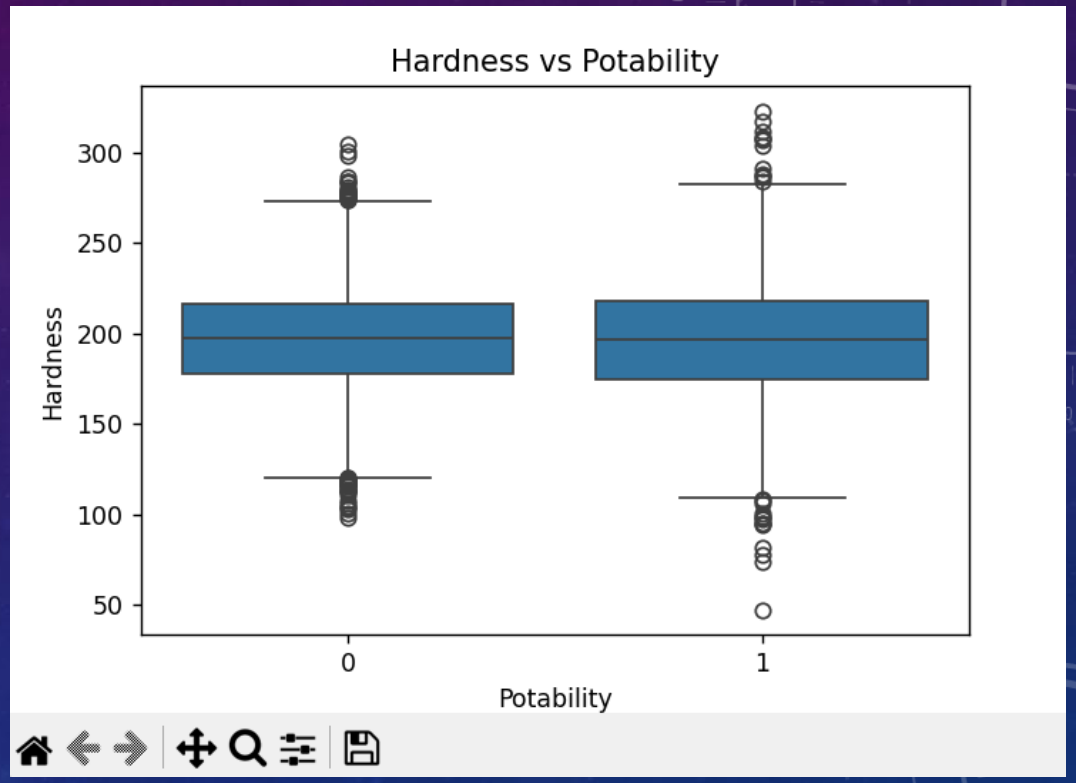


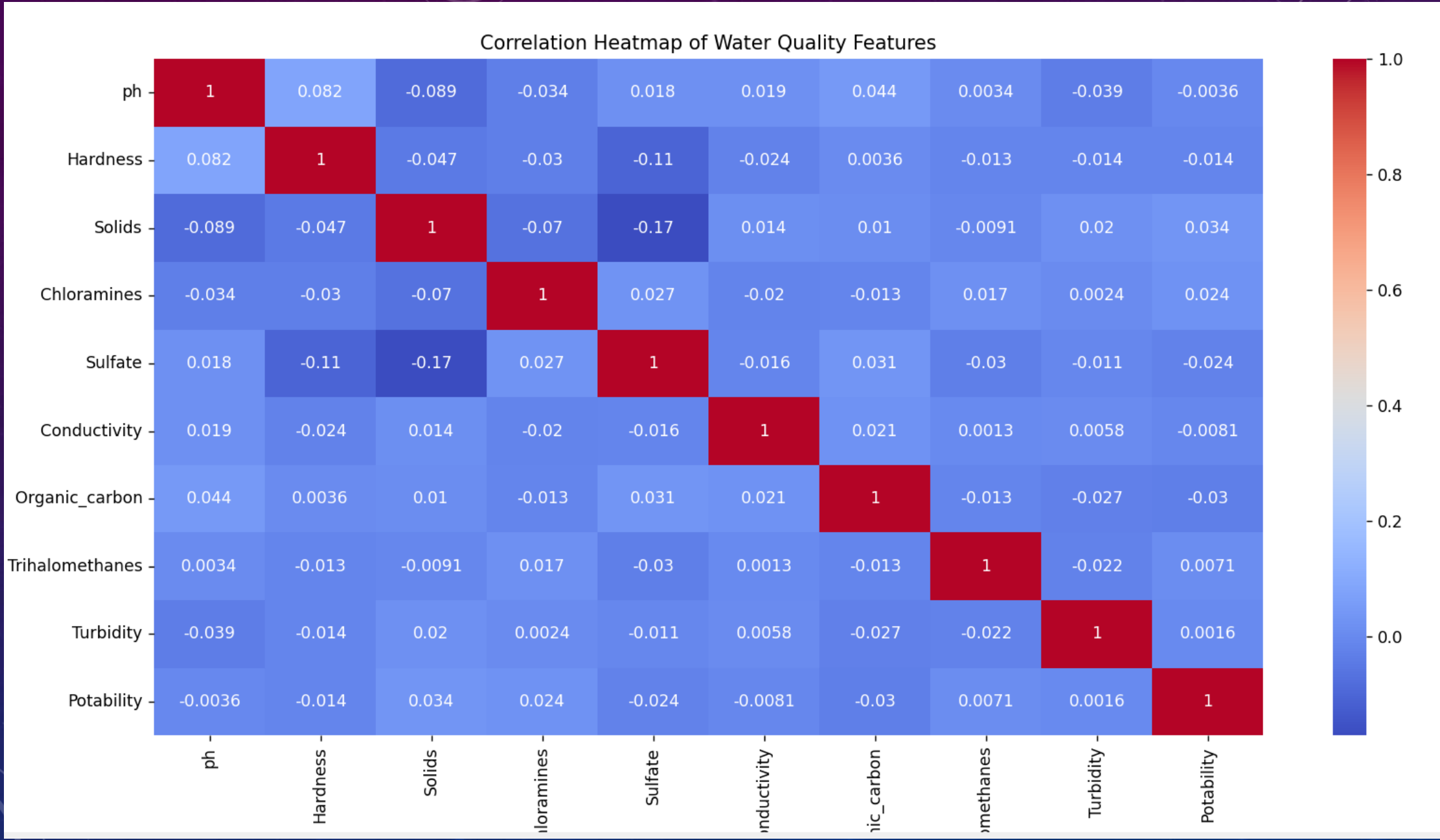


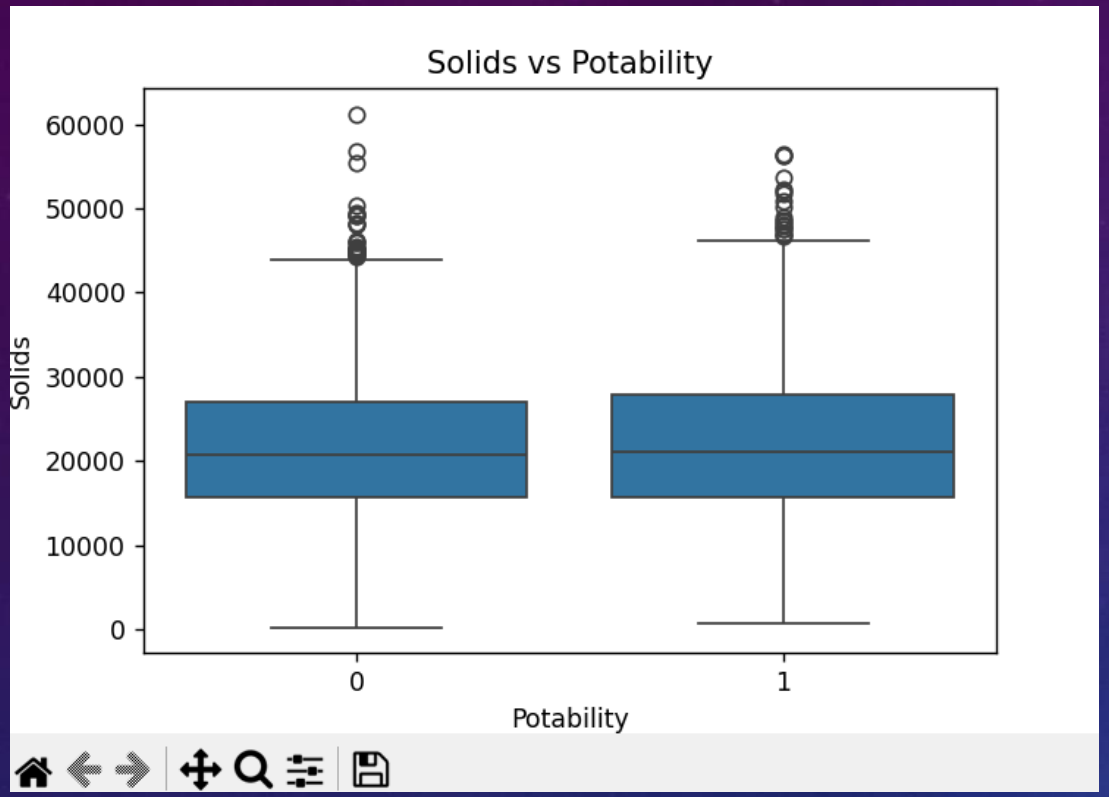

Distribution of Hardness
Frequency
Hardness
0
50
100
150
200
250
300
350
400
50
100
150
200
250
300
(x, y) = (158.0, 437.)

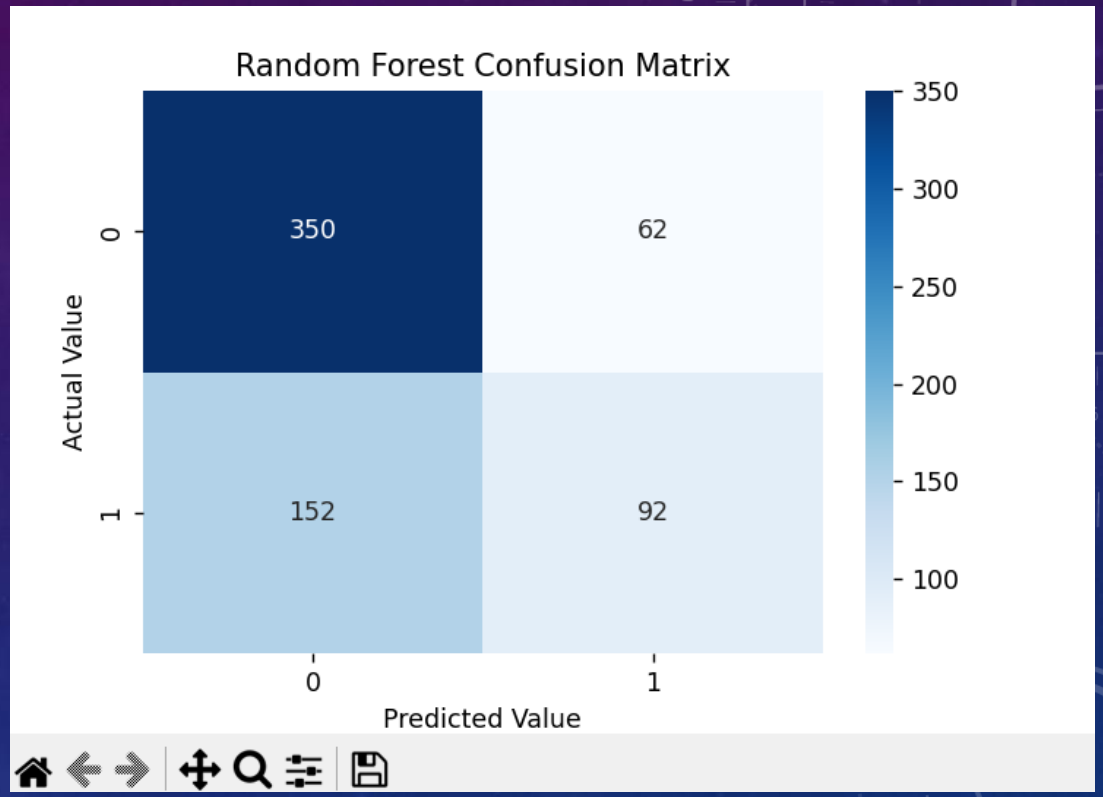

Distribution of Solids
Frequency
Solids
0
50
100
150
200
250
300
350
0
10000
20000
30000
40000
50000
60000
(x, y) = (1.71e+04, 339.8)

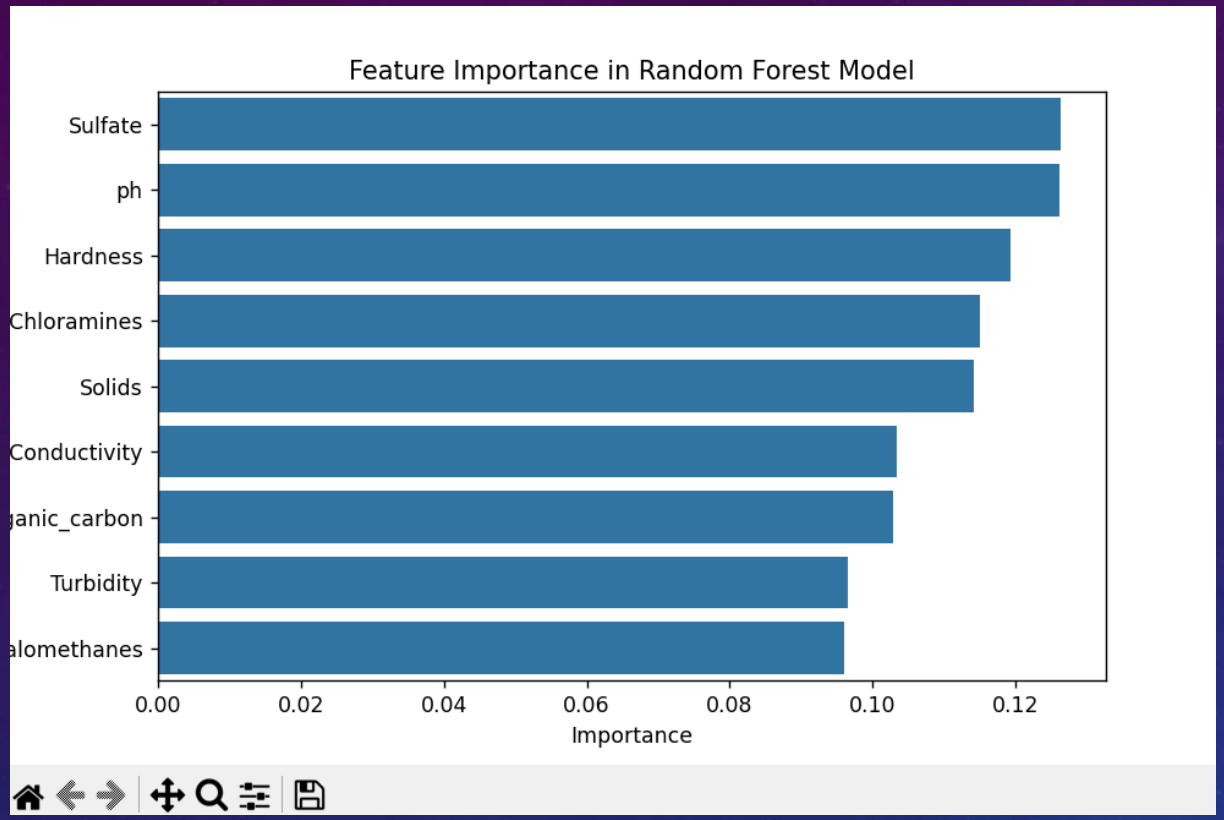

pH vs Potability
pH
14
12
10
8
6
4
2
0
0
1
Potability
(x, y) = (1, 13.94)

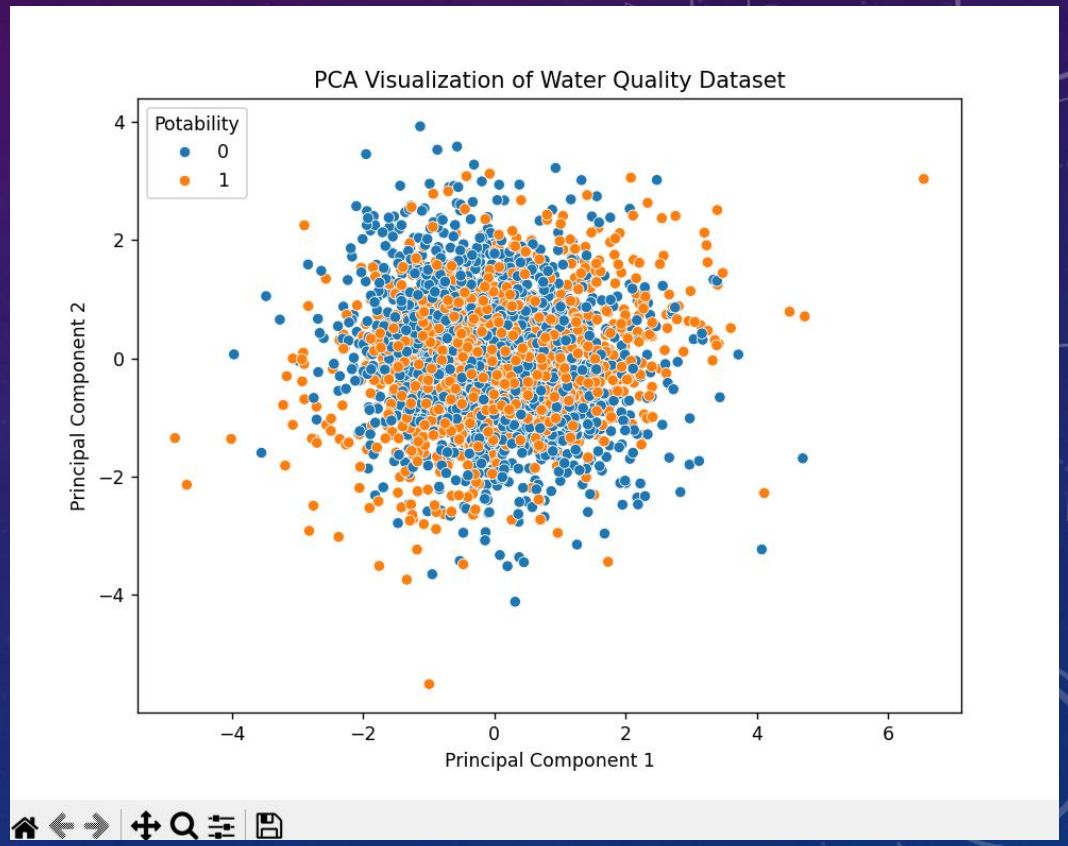

Hardness vs Potability
Hardness
300
250
200
150
100
50
0
1
Potability


Correlation Heatmap of Water Quality Features
ph 1 0.082 -0.089 -0.034 0.018 0.019 0.044 0.0034 -0.039 -0.0036
Hardness 0.082 1 -0.047 -0.03 -0.11 -0.024 0.0036 -0.013 -0.014 -0.014
Solids -0.089 -0.047 1 -0.07 -0.17 0.014 0.01 -0.0091 0.02 0.034
Chloramines -0.034 -0.03 -0.07 1 0.027 -0.02 -0.013 0.017 0.0024 0.024
Sulfate 0.018 -0.11 -0.17 0.027 1 -0.016 0.031 -0.03 -0.011 -0.024
Conductivity 0.019 -0.024 0.014 -0.02 -0.016 1 0.021 0.0013 0.0058 -0.0081
Organic_carbon 0.044 0.0036 0.01 -0.013 0.031 0.021 1 -0.013 -0.027 -0.03
Trihalomethanes 0.0034 -0.013 -0.0091 0.017 -0.03 0.0013 -0.013 1 -0.022 0.0071
Turbidity -0.039 -0.014 0.02 0.0024 -0.011 0.0058 -0.027 -0.022 1 0.0016
Potability -0.0036 -0.014 0.034 0.024 -0.024 -0.0081 -0.03 0.0071 0.0016 1
ph
Hardness
Solids
Sulfate
Turbidity
Potability
1.0
0.8
0.6
0.4
0.2
0.0


Solids vs Potability
Solids
60000
50000
40000
30000
20000
10000
0
0
1
Potability


Random Forest Confusion Matrix
Actual Value
0
1
350
62
152
92
0
1
Predicted Value
350
300
250
200
150
100


Feature Importance in Random Forest Model
Sulfate
ph
Hardness
Chloramines
Solids
Conductivity
ganic_carbon
Turbidity
alomethanes
0.00
0.02
0.04
0.06
0.08
0.10
0.12
Importance


PCA Visualization of Water Quality Dataset
Potability
0
1
Principal Component 2
Principal Component 1
4
2
0
−2
−4
−4
−2
0
2
4
6

Future Scope

- Use more advanced models (XGBoost, Neural Networks)
- Deploy as web application
- Real-time water monitoring system

Conclusion

- Successfully analyzed water quality data
- Built a prediction model using Random Forest
- Achieved good accuracy
- Identified important water parameters
- PCA improved visualization

THANK YOU!!

AI-ML PROJECT

Water Quality Analysis and Potability Prediction Using Machine Learning

Aim : The aim of my project is to analyze water quality data and predict whether water is potable or not potable. This is a supervised machine learning classification project because the output is a class, either 0 for not potable or 1 for potable. I used the water_potability.csv dataset, which contains features like pH, Hardness, Solids, Chloramines, Sulfate, Conductivity, Organic Carbon, Trihalomethanes, and Turbidity.

First, I loaded the dataset using Pandas and performed exploratory data analysis using countplot, histogram, boxplot, and heatmap. Then I handled missing values using median imputation because median is less affected by outliers. After that, I separated the input features and target variable. I scaled the features using StandardScaler because PCA was used later.

Then I split the dataset into 80% training and 20% testing data. I used Random Forest Classifier as the main algorithm because it works well for classification, handles non-linear patterns, reduces overfitting, and gives feature importance. After training the model, I evaluated it using accuracy score, confusion matrix, and classification report. I also used feature importance to identify important water quality parameters and PCA to visualize the dataset in two dimensions.

The final output of the project is whether a water sample is safe to drink or not safe to drink.

CODE:

```
# ============================================================
# WATER QUALITY ANALYSIS AND POTABILITY PREDICTION PROJECT
# ============================================================

# ---------------- IMPORTING LIBRARIES ----------------
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, confusion_matrix,
classification_report
```

```python
from sklearn.decomposition import PCA


# ============================================================
# LOADING DATASET
# ============================================================

df = pd.read_csv("water_potability.csv")

print("\n========== FIRST 5 ROWS ==========")
print(df.head())

print("\n========== DATASET INFORMATION ==========")
print(df.info())

print("\n========== STATISTICAL SUMMARY ==========")
print(df.describe())

print("\n========== DATASET SHAPE ==========")
print("Rows and Columns:", df.shape)

print("\n========== MISSING VALUES BEFORE HANDLING ==========")
print(df.isnull().sum())


# ============================================================
# EXPLORATORY DATA ANALYSIS
# ============================================================

# ---------------- UNIVARIATE ANALYSIS ----------------

# Count of potable and non-potable water
plt.figure(figsize=(6, 4))
sns.countplot(x="Potability", data=df)
plt.title("Count of Potable and Non-Potable Water")
plt.xlabel("Potability")
plt.ylabel("Count")
plt.show()

# Histogram of pH
plt.figure(figsize=(6, 4))
sns.histplot(df["ph"], bins=30, kde=True)
plt.title("Distribution of pH")
plt.xlabel("pH Value")
plt.ylabel("Frequency")
plt.show()

# Histogram of Hardness
plt.figure(figsize=(6, 4))
```

```python
sns.histplot(df["Hardness"], bins=30, kde=True)
plt.title("Distribution of Hardness")
plt.xlabel("Hardness")
plt.ylabel("Frequency")
plt.show()

# Histogram of Solids
plt.figure(figsize=(6, 4))
sns.histplot(df["Solids"], bins=30, kde=True)
plt.title("Distribution of Solids")
plt.xlabel("Solids")
plt.ylabel("Frequency")
plt.show()


# ---------------- BIVARIATE ANALYSIS ----------------

# pH vs Potability
plt.figure(figsize=(6, 4))
sns.boxplot(x="Potability", y="ph", data=df)
plt.title("pH vs Potability")
plt.xlabel("Potability")
plt.ylabel("pH")
plt.show()

# Hardness vs Potability
plt.figure(figsize=(6, 4))
sns.boxplot(x="Potability", y="Hardness", data=df)
plt.title("Hardness vs Potability")
plt.xlabel("Potability")
plt.ylabel("Hardness")
plt.show()

# Solids vs Potability
plt.figure(figsize=(6, 4))
sns.boxplot(x="Potability", y="Solids", data=df)
plt.title("Solids vs Potability")
plt.xlabel("Potability")
plt.ylabel("Solids")
plt.show()


# ---------------- MULTIVARIATE ANALYSIS ----------------

plt.figure(figsize=(10, 7))
sns.heatmap(df.corr(), annot=True, cmap="coolwarm")
plt.title("Correlation Heatmap of Water Quality Features")
plt.show()
```

```python
# ============================================================
# MISSING VALUE HANDLING
# ============================================================

print("\n========== HANDLING MISSING VALUES ==========")

# Making a copy of original dataset
df_cleaned = df.copy()

# Filling missing numerical values using median
for column in df_cleaned.columns:
    if df_cleaned[column].isnull().sum() > 0:
        median_value = df_cleaned[column].median()
        df_cleaned[column] = df_cleaned[column].fillna(median_value)

print("\n========== MISSING VALUES AFTER HANDLING ==========")
print(df_cleaned.isnull().sum())

print("\nTotal missing values after handling:")
print(df_cleaned.isnull().sum().sum())


# ============================================================
# FEATURE SELECTION
# ============================================================

# X contains input features
X = df_cleaned.drop("Potability", axis=1)

# y contains target/output column
y = df_cleaned["Potability"]


# ============================================================
# FEATURE SCALING
# ============================================================

scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

print("\n========== FEATURE SCALING COMPLETED ==========")
print("Features are standardized using StandardScaler.")


# ============================================================
# TRAIN TEST SPLIT
# ============================================================

X_train, X_test, y_train, y_test = train_test_split(
```

```python
    X_scaled,
    y,
    test_size=0.2,
    random_state=42
)

print("\n========== TRAIN TEST SPLIT ==========")
print("Training data size:", X_train.shape)
print("Testing data size:", X_test.shape)


# ============================================================
# MODEL TRAINING - RANDOM FOREST
# ============================================================

model = RandomForestClassifier(
    n_estimators=100,
    random_state=42
)

model.fit(X_train, y_train)

print("\n========== MODEL TRAINING COMPLETED ==========")
print("Random Forest Classifier model has been trained.")


# ============================================================
# MODEL PREDICTION
# ============================================================

y_pred = model.predict(X_test)


# ============================================================
# MODEL EVALUATION
# ============================================================

accuracy = accuracy_score(y_test, y_pred)

print("\n========== RANDOM FOREST MODEL ACCURACY ==========")
print("Accuracy:", accuracy * 100, "%")

print("\n========== CONFUSION MATRIX ==========")
print(confusion_matrix(y_test, y_pred))

print("\n========== CLASSIFICATION REPORT ==========")
print(classification_report(y_test, y_pred))
```

```python
# Confusion Matrix Graph
plt.figure(figsize=(6, 4))
sns.heatmap(confusion_matrix(y_test, y_pred), annot=True, fmt="d",
cmap="Blues")
plt.title("Random Forest Confusion Matrix")
plt.xlabel("Predicted Value")
plt.ylabel("Actual Value")
plt.show()


# ============================================================
# FEATURE IMPORTANCE
# ============================================================

feature_importance = pd.DataFrame({
    "Feature": X.columns,
    "Importance": model.feature_importances_
})

feature_importance = feature_importance.sort_values(by="Importance",
ascending=False)

print("\n========== FEATURE IMPORTANCE ==========")
print(feature_importance)

plt.figure(figsize=(8, 5))
sns.barplot(x="Importance", y="Feature", data=feature_importance)
plt.title("Feature Importance in Random Forest Model")
plt.xlabel("Importance")
plt.ylabel("Water Quality Features")
plt.show()


# ============================================================
# PCA - DIMENSIONALITY REDUCTION
# ============================================================

pca = PCA(n_components=2)
X_pca = pca.fit_transform(X_scaled)

pca_df = pd.DataFrame(data=X_pca, columns=["PC1", "PC2"])
pca_df["Potability"] = y

print("\n========== PCA EXPLAINED VARIANCE RATIO ==========")
print(pca.explained_variance_ratio_)

plt.figure(figsize=(8, 6))
sns.scatterplot(x="PC1", y="PC2", hue="Potability", data=pca_df)
plt.title("PCA Visualization of Water Quality Dataset")
```

```
plt.xlabel("Principal Component 1")
plt.ylabel("Principal Component 2")
plt.show()


# ============================================================
# SAMPLE PREDICTION
# ============================================================

print("\n========== SAMPLE PREDICTION ==========")

sample = X_test[0].reshape(1, -1)
prediction = model.predict(sample)

if prediction[0] == 1:
    print("The sample water is predicted as: POTABLE / SAFE TO DRINK")
else:
    print("The sample water is predicted as: NOT POTABLE / NOT SAFE TO DRINK")


# ============================================================
# PROJECT CONCLUSION
# ============================================================

print("\n========== CONCLUSION ==========")
print("This project analyzed water quality data using EDA and preprocessing
techniques.")
print("Missing values were handled using median imputation.")
print("Features were scaled using StandardScaler.")
print("A Random Forest Classifier model was trained to predict water
potability.")
print("Feature importance was used to identify important water quality
parameters.")
print("PCA was applied to reduce the dataset into two principal components for
visualization.")
print("Random Forest performed better because it can learn non-linear patterns
from multiple water quality features.")
```

Output:

========== FIRST 5 ROWS ==========

ph Hardness Solids Chloramines Sulfate Conductivity Organic_carbon Trihalomethanes Turbidity Potability

```
0      NaN  204.890455  20791.318981     7.300212  368.516441    564.308654
10.379783       86.990970   2.963135          0

1  3.716080  129.422921  18630.057858     6.635246        NaN    592.885359
15.180013       56.329076   4.500656          0

2  8.099124  224.236259  19909.541732     9.275884        NaN    418.606213
16.868637       66.420093   3.055934          0

3  8.316766  214.373394  22018.417441     8.059332  356.886136    363.266516
18.436524      100.341674   4.628771          0

4  9.092223  181.101509  17978.986339     6.546600  310.135738    398.410813
11.558279       31.997993   4.075075          0


========== DATASET INFORMATION ==========

<class 'pandas.DataFrame'>

RangeIndex: 3276 entries, 0 to 3275

Data columns (total 10 columns):

 #   Column          Non-Null Count  Dtype

---  ------          --------------  -----

 0   ph              2785 non-null   float64

 1   Hardness        3276 non-null   float64

 2   Solids          3276 non-null   float64

 3   Chloramines     3276 non-null   float64

 4   Sulfate         2495 non-null   float64

 5   Conductivity    3276 non-null   float64

 6   Organic_carbon  3276 non-null   float64

 7   Trihalomethanes 3114 non-null   float64

 8   Turbidity       3276 non-null   float64

 9   Potability      3276 non-null   int64

dtypes: float64(9), int64(1)
```

memory usage: 256.1 KB

None

========== STATISTICAL SUMMARY ==========

	ph	Hardness	Solids	Chloramines	Sulfate	Conductivity	Organic_carbon	Trihalomethanes	Turbidity	Potability
count	2785.000000	3276.000000	3276.000000	3276.000000	2495.000000	3276.000000	3276.000000	3114.000000	3276.000000	3276.000000
mean	7.080795	196.369496	22014.092526	7.122277	333.775777	426.205111	14.284970	66.396293	3.966786	0.390110
std	1.594320	32.879761	8768.570828	1.583085	41.416840	80.824064	3.308162	16.175008	0.780382	0.487849
min	0.000000	47.432000	320.942611	0.352000	129.000000	181.483754	2.200000	0.738000	1.450000	0.000000
25%	6.093092	176.850538	15666.690297	6.127421	307.699498	365.734414	12.065801	55.844536	3.439711	0.000000
50%	7.036752	196.967627	20927.833607	7.130299	333.073546	421.884968	14.218338	66.622485	3.955028	0.000000
75%	8.062066	216.667456	27332.762127	8.114887	359.950170	481.792304	16.557652	77.337473	4.500320	1.000000
max	14.000000	323.124000	61227.196008	13.127000	481.030642	753.342620	28.300000	124.000000	6.739000	1.000000

========== DATASET SHAPE ==========

Rows and Columns: (3276, 10)

========== MISSING VALUES BEFORE HANDLING ==========

ph 491

Hardness 0

```
Solids             0
Chloramines        0
Sulfate          781
Conductivity       0
Organic_carbon     0
Trihalomethanes  162
Turbidity          0
Potability         0
dtype: int64

========== HANDLING MISSING VALUES ==========

========== MISSING VALUES AFTER HANDLING ==========
ph               0
Hardness         0
Solids           0
Chloramines      0
Sulfate          0
Conductivity     0
Organic_carbon   0
Trihalomethanes  0
Turbidity        0
Potability       0
dtype: int64

Total missing values after handling:
```

```
0

========== FEATURE SCALING COMPLETED ==========
Features are standardized using StandardScaler.

========== TRAIN TEST SPLIT ==========
Training data size: (2620, 9)
Testing data size: (656, 9)

========== MODEL TRAINING COMPLETED ==========
Random Forest Classifier model has been trained.

========== RANDOM FOREST MODEL ACCURACY ==========
Accuracy: 67.3780487804878 %

========== CONFUSION MATRIX ==========
[[350  62]
 [152  92]]

========== CLASSIFICATION REPORT ==========
              precision    recall  f1-score   support

           0       0.70      0.85      0.77       412
           1       0.60      0.38      0.46       244

    accuracy                           0.67       656
```

macro avg	0.65	0.61	0.61	656
weighted avg	0.66	0.67	0.65	656

========== FEATURE IMPORTANCE ==========

	Feature	Importance
4	Sulfate	0.126330
0	ph	0.126106
1	Hardness	0.119346
3	Chloramines	0.115096
2	Solids	0.114246
5	Conductivity	0.103434
6	Organic_carbon	0.102947
8	Turbidity	0.096463
7	Trihalomethanes	0.096031

========== PCA EXPLAINED VARIANCE RATIO ==========

[0.13337387 0.12632612]

========== SAMPLE PREDICTION ==========

The sample water is predicted as: NOT POTABLE / NOT SAFE TO DRINK

========== CONCLUSION ==========

This project analyzed water quality data using EDA and preprocessing techniques.

Missing values were handled using median imputation.

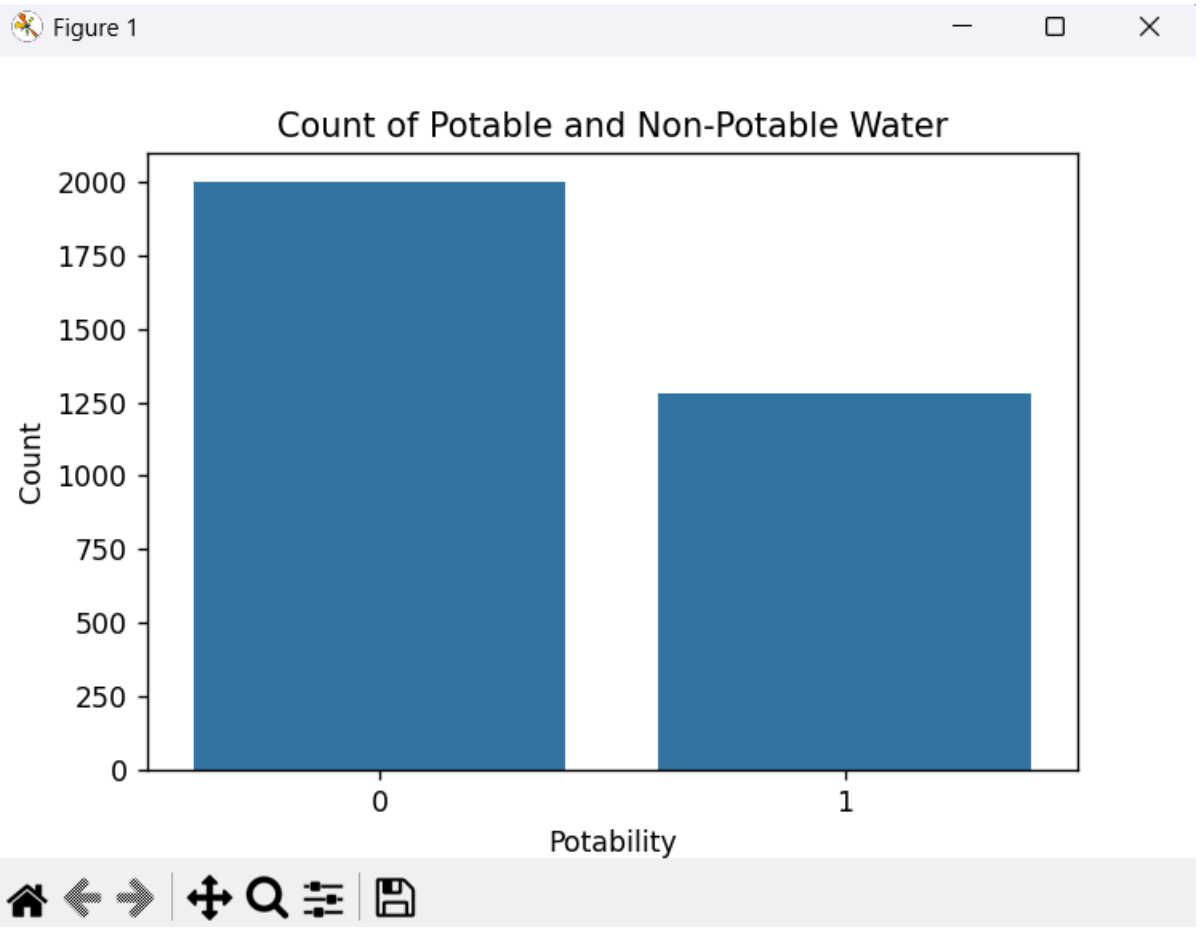
Features were scaled using StandardScaler.

A Random Forest Classifier model was trained to predict water potability.

Feature importance was used to identify important water quality parameters.

PCA was applied to reduce the dataset into two principal components for visualization.

Random Forest performed better because it can learn non-linear patterns from multiple water quality features.



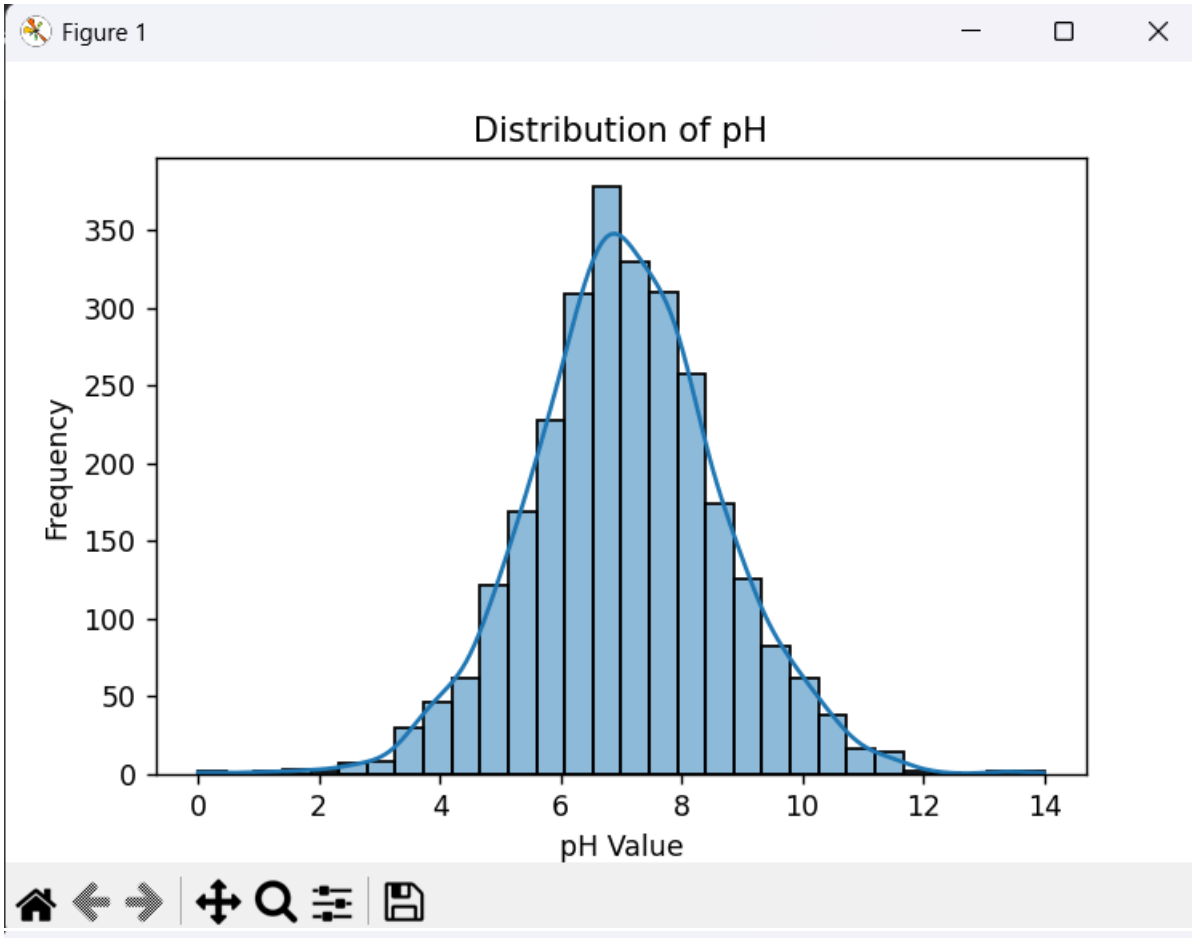


Figure 1
Distribution of pH
Frequency
0
50
100
150
200
250
300
350
0
2
4
6
8
10
12
14
pH Value

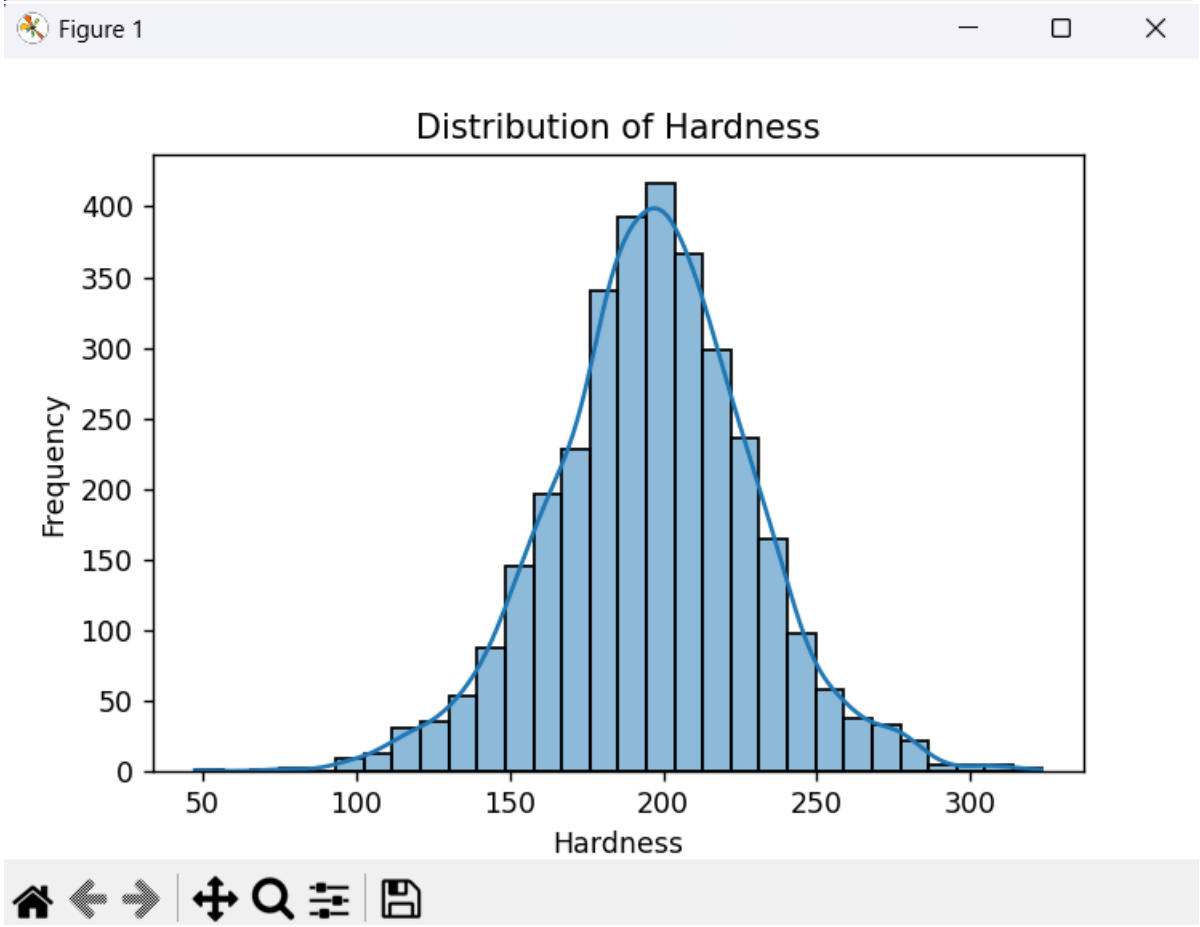


Figure 1
Distribution of Hardness
Frequency
0
50
100
150
200
250
300
350
400
50
100
150
200
250
300
Hardness

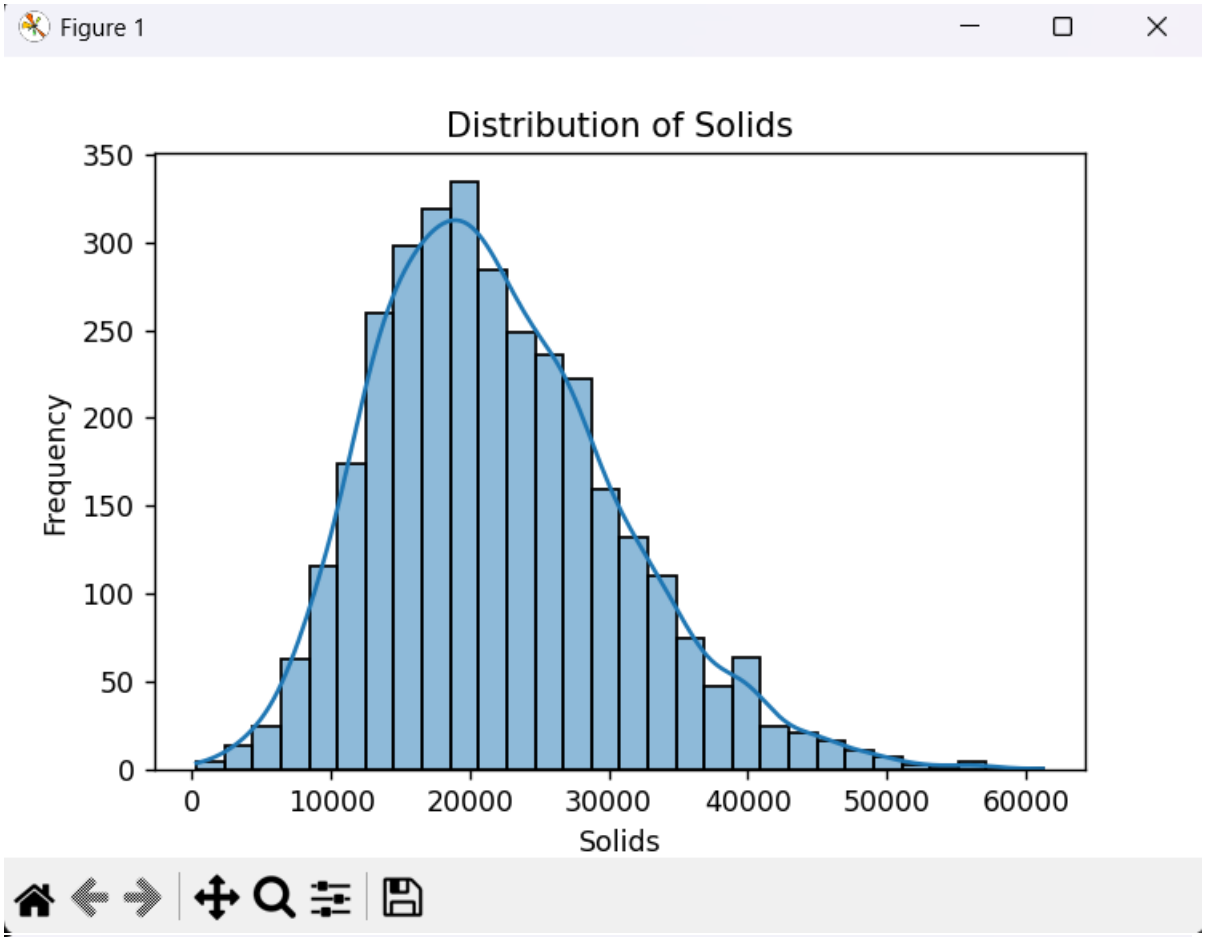


Figure 1
Distribution of Solids
Frequency
Solids
0
50
100
150
200
250
300
350
0
10000
20000
30000
40000
50000
60000

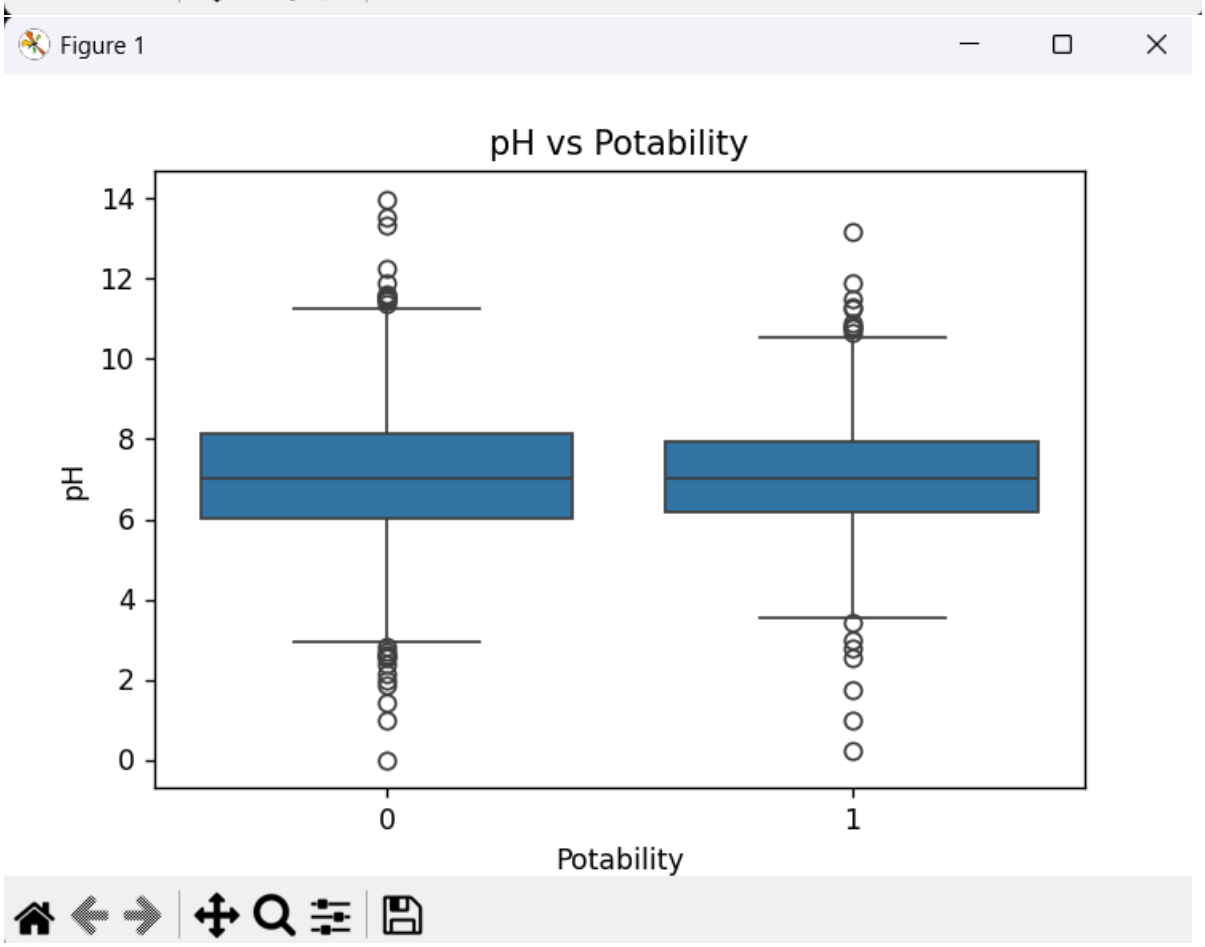


Figure 1
pH vs Potability
pH
Potability
0
2
4
6
8
10
12
14
0
1

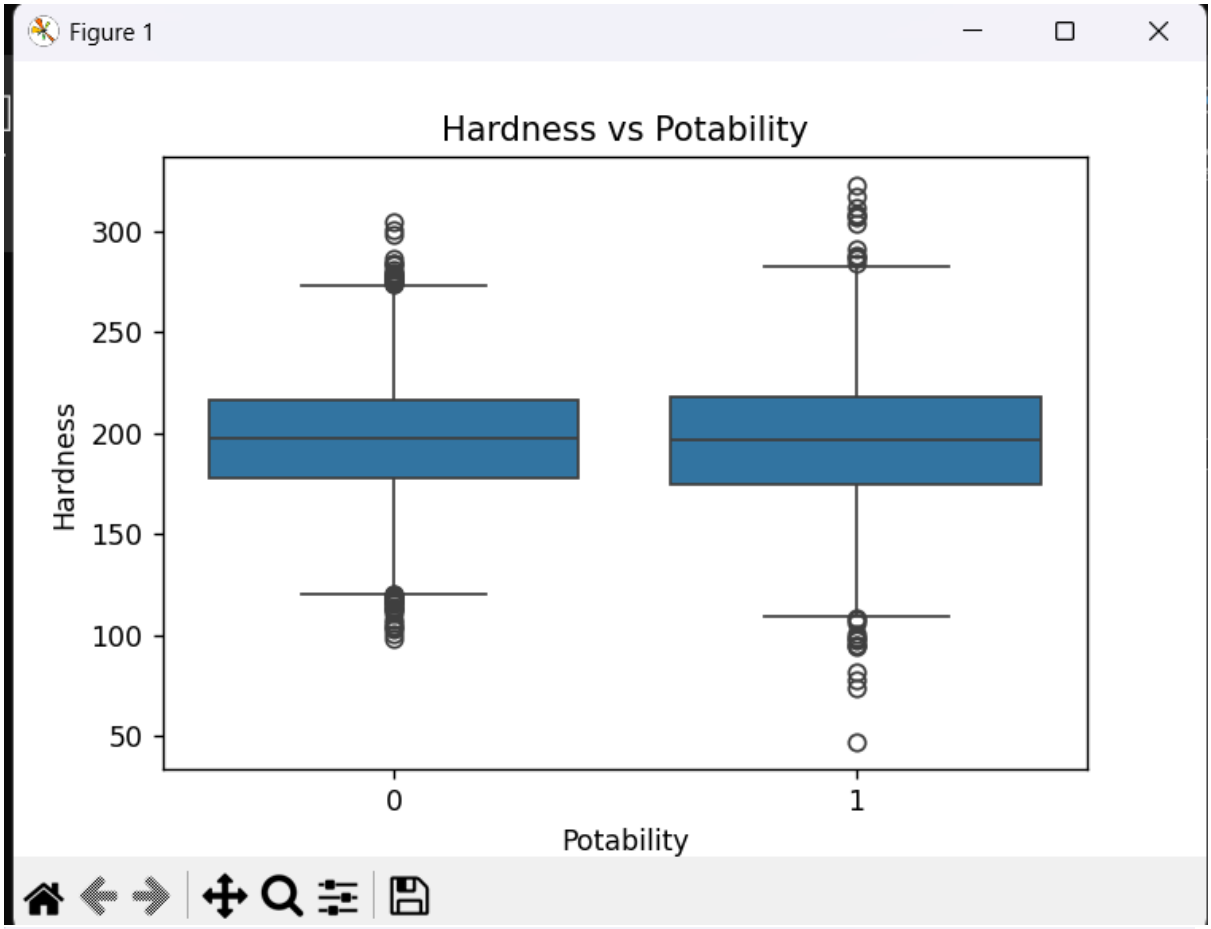


Figure 1
Hardness vs Potability
Hardness
300
250
200
150
100
50
0
1
Potability

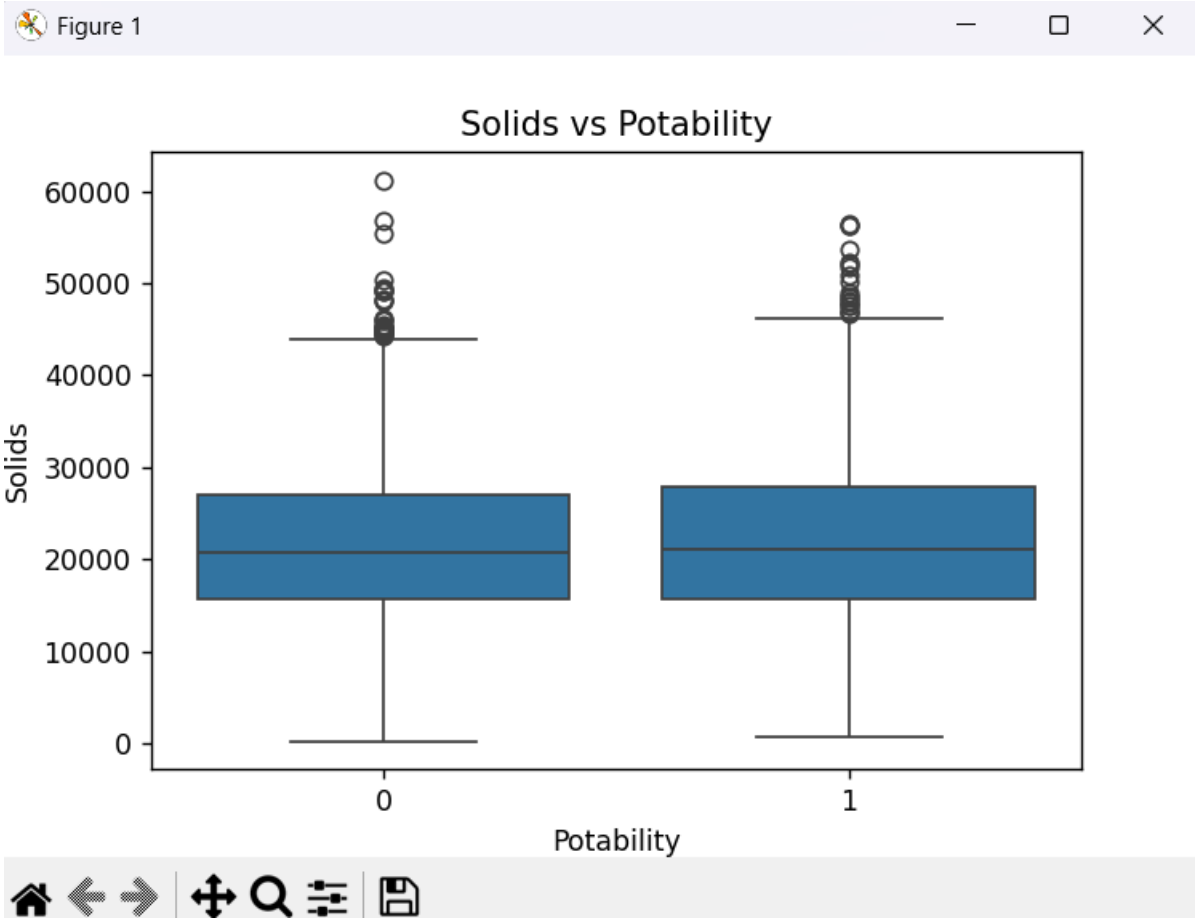


Figure 1
Solids vs Potability
Solids
60000
50000
40000
30000
20000
10000
0
0
1
Potability

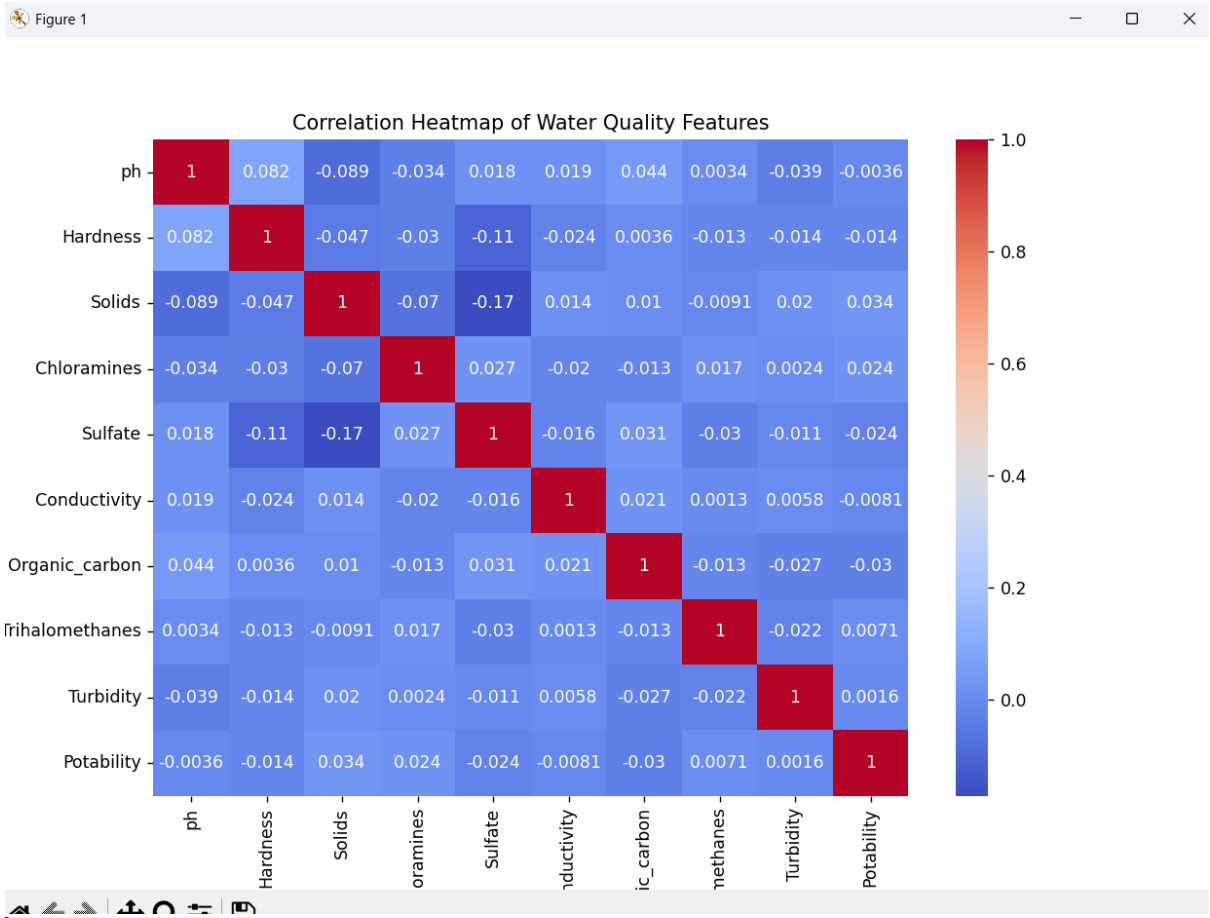


Figure 1
Correlation Heatmap of Water Quality Features
ph 1 0.082 -0.089 -0.034 0.018 0.019 0.044 0.0034 -0.039 -0.0036
Hardness 0.082 1 -0.047 -0.03 -0.11 -0.024 0.0036 -0.013 -0.014 -0.014
Solids -0.089 -0.047 1 -0.07 -0.17 0.014 0.01 -0.0091 0.02 0.034
Chloramines -0.034 -0.03 -0.07 1 0.027 -0.02 -0.013 0.017 0.0024 0.024
Sulfate 0.018 -0.11 -0.17 0.027 1 -0.016 0.031 -0.03 -0.011 -0.024
Conductivity 0.019 -0.024 0.014 -0.02 -0.016 1 0.021 0.0013 0.0058 -0.0081
Organic_carbon 0.044 0.0036 0.01 -0.013 0.031 0.021 1 -0.013 -0.027 -0.03
Trihalomethanes 0.0034 -0.013 -0.0091 0.017 -0.03 0.0013 -0.013 1 -0.022 0.0071
Turbidity -0.039 -0.014 0.02 0.0024 -0.011 0.0058 -0.027 -0.022 1 0.0016
Potability -0.0036 -0.014 0.034 0.024 -0.024 -0.0081 -0.03 0.0071 0.0016 1
ph
Hardness
Solids
oramines
Sulfate
nductivity
ic_carbon
methanes
Turbidity
Potability
1.0
0.8
0.6
0.4
0.2
0.0

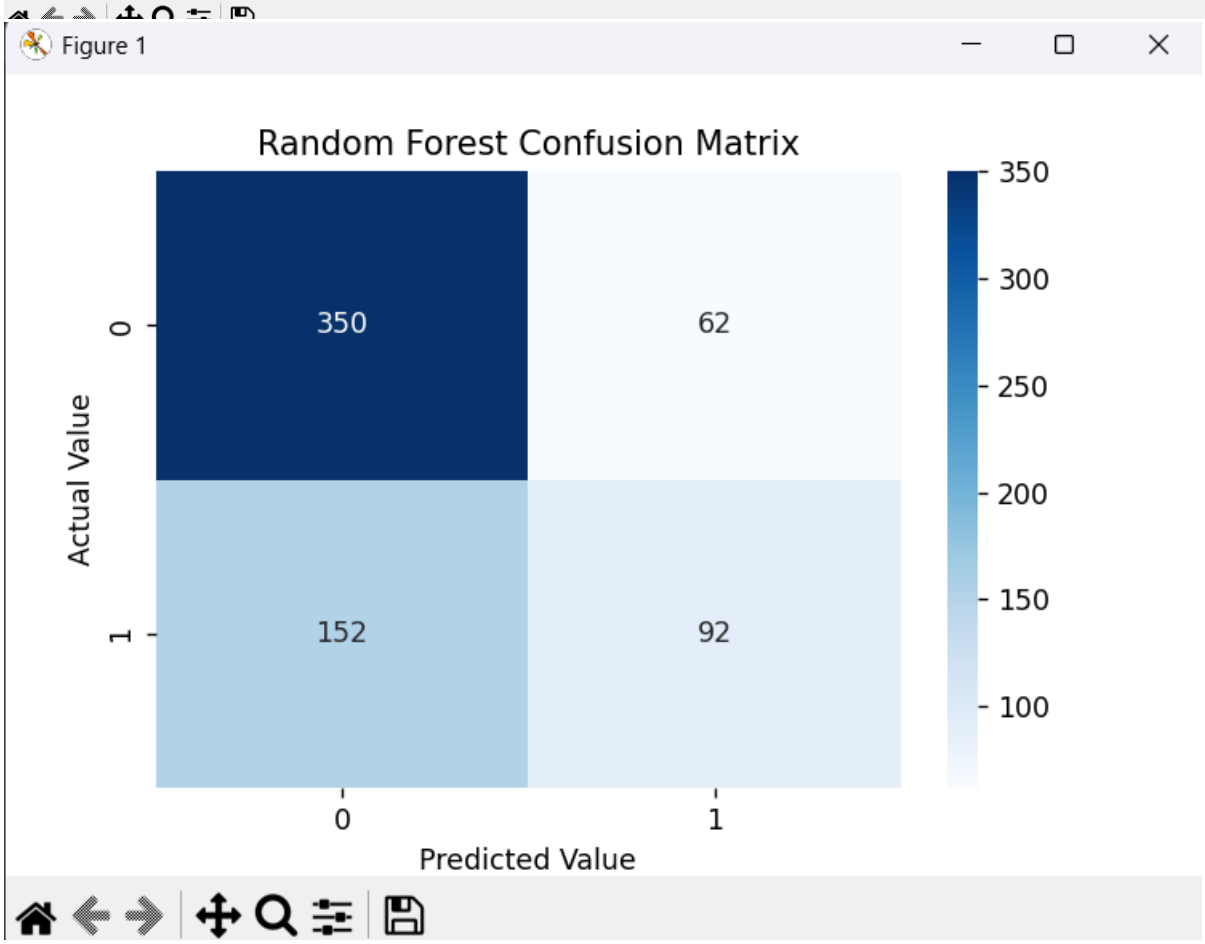


Figure 1
Random Forest Confusion Matrix
350
62
152
92
0
1
Actual Value
0
1
Predicted Value
350
300
250
200
150
100

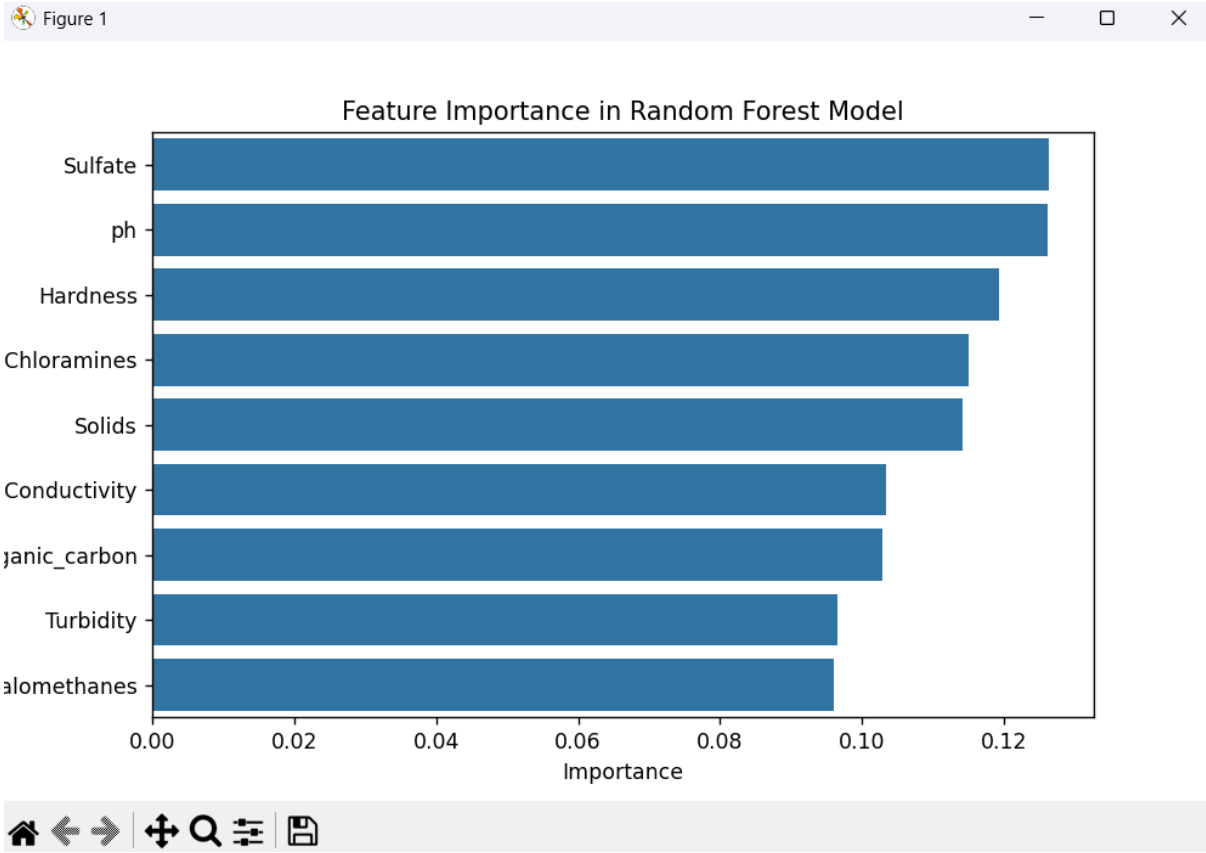


Figure 1
Feature Importance in Random Forest Model
Sulfate
ph
Hardness
Chloramines
Solids
Conductivity
ganic_carbon
Turbidity
alomethanes
0.00
0.02
0.04
0.06
0.08
0.10
0.12
Importance

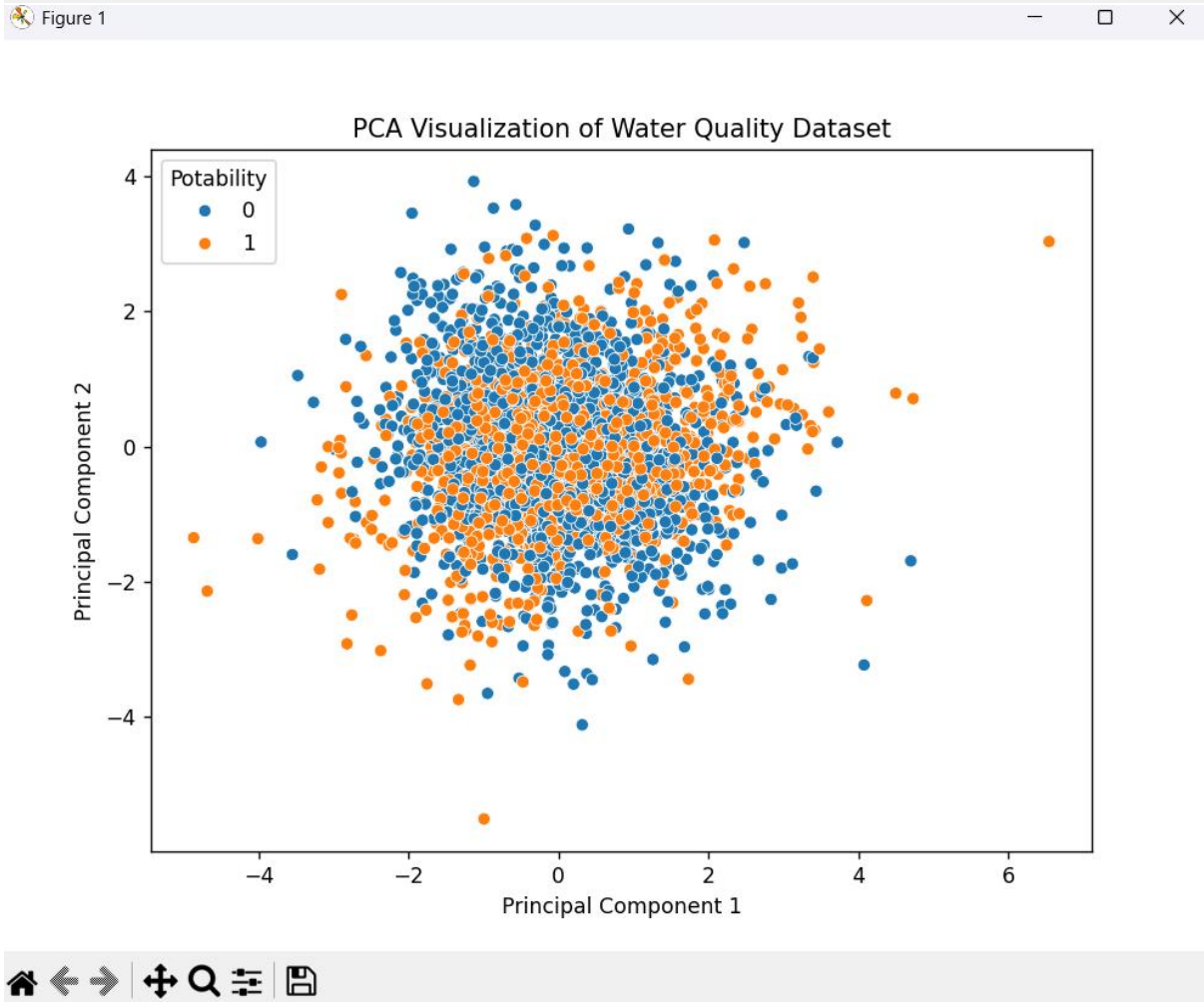


Figure 1
PCA Visualization of Water Quality Dataset
Potability
0
1
4
2
0
−2
−4
Principal Component 2
−4
−2
0
2
4
6
Principal Component 1